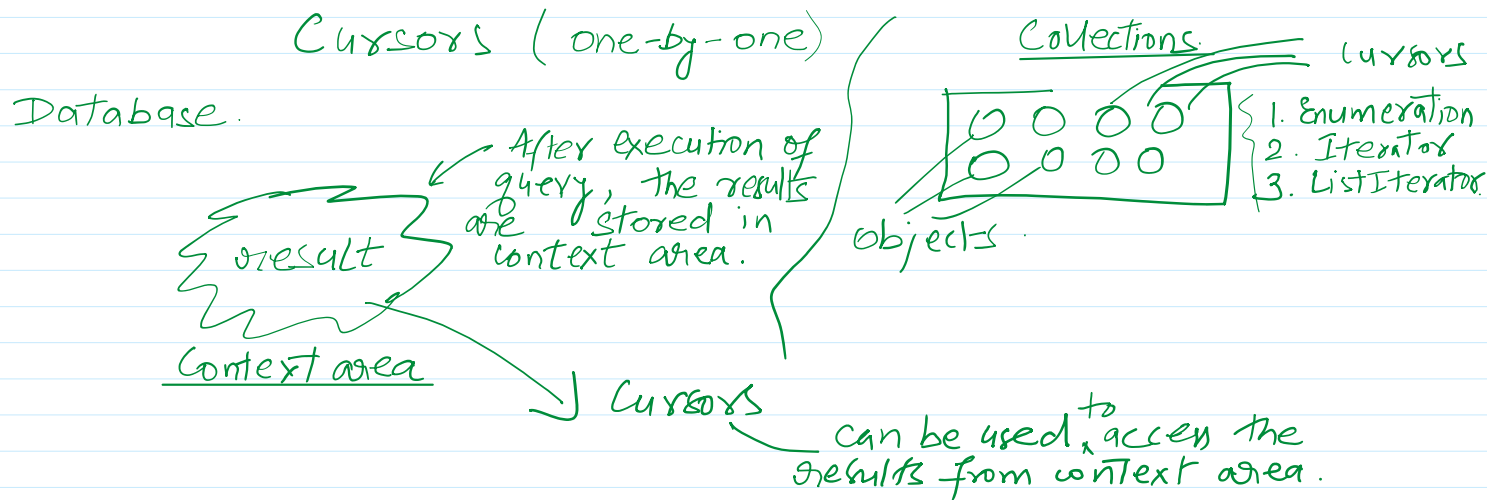




Two types of cursors:

1. Implicit Cursors
2. Explicit Cursors

Some cursors created automatically by database softwares to hold the results of SQL queries. Whenever any query is executed, it executes implicit cursor to hold the results of query.

The cursors which are created by the programmers, to hold the result of query is called explicit cursor.

1. **SYS-REFCURSOR** :- it is used to hold the result of select query.
2. **%ROWCOUNT** :- the result of non-select query, insert, update, delete query.
3. **%FOUND** :- check whether any rows effected because of non-select query.

STORED PROCEDURE CURSOR Application.

```

create or replace procedure getEmpInfo (
    emps OUT SYS-REFCURSOR) as
BEGIN
    open emps for
    select * from employee;
END.
```

```

CREATE OR REPLACE PROCEDURE GETALLEMPINFO(
    emps OUT SYS_REFCURSOR
)AS
BEGIN
    open emps for
    select * from employee;
END GETALLEMPINFO;

```

Output Variables - Log

Variable	Value																																								
EMPS	<table><tr><th>EMP_ID</th><th>NAME</th><th>DEPT_ID</th><th>BASIC_SAL...</th></tr><tr><td>E006</td><td>rashika</td><td>0003</td><td>46500</td></tr><tr><td>E009</td><td>Amar</td><td>D003</td><td>43800</td></tr><tr><td>E010</td><td>Frank</td><td>IT</td><td>67000</td></tr><tr><td>E011</td><td>Rickard</td><td>Hardware</td><td>70000</td></tr><tr><td>E012</td><td>Roberto</td><td>Hardware</td><td>70100</td></tr><tr><td>E013</td><td>Sakshi</td><td>IT</td><td>67800</td></tr><tr><td>E014</td><td>Eeshan</td><td>D001</td><td>45600</td></tr><tr><td>E015</td><td>Rajveer</td><td>D002</td><td>71200</td></tr><tr><td>E016</td><td>Jagdish</td><td>D003</td><td>46800</td></tr></table>	EMP_ID	NAME	DEPT_ID	BASIC_SAL...	E006	rashika	0003	46500	E009	Amar	D003	43800	E010	Frank	IT	67000	E011	Rickard	Hardware	70000	E012	Roberto	Hardware	70100	E013	Sakshi	IT	67800	E014	Eeshan	D001	45600	E015	Rajveer	D002	71200	E016	Jagdish	D003	46800
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E016	Jagdish	D003	46800																																						

Messages | Logging Page | Statements | Running: SatyendraConnection.jpr | Output Variables

```

package MyJDBCOne;
import java.sql.CallableStatement;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import oracle.jdbc.OracleTypes;

public class StoredProcedureDemo2 {
    public static void main(String[] args) {
        String dbUrl = "jdbc:oracle:thin:@//localhost:1521/FREE";
        String username = "sys as sysdba";
        String password = "Calcutta1983";
        Connection con;
        try {
            con = DriverManager.getConnection(dbUrl, username, password);
            CallableStatement cst = con.prepareCall("{call getallempinfo(?)}");
            cst.registerOutParameter(1, OracleTypes.CURSOR);
            cst.execute();
            ResultSet rs = (ResultSet) cst.getObject(1);
            boolean flag = false;
            System.out.println("Emp_id\tName\tdept_id\tbasic_salary");
            System.out.println("-----");
            while (rs.next()) {
                flag = true;
                System.out.println(rs.getString(1) + "\t" +
                    rs.getString(2) + "\t" +
                    rs.getString(3) + "\t\t" +
                    rs.getFloat(4));
            }
            if (!flag) {
                System.out.println("No records available");
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}

```

Output:

Emp_id	Name	dept_id	basic_salary
E001	Pratik Jain	D001	56000.0
E002	Naveen Singh	D001	67800.0
E003	Sangita Mishra	D004	45700.0
E004	Ajay Dewan	D002	34000.0
E005	Bali Agarwal	D002	35600.0
E006	Gargi	IT	70000.0
E007	Garvit	D004	45000.0
E008	Yashika	D003	46500.0
E009	Amar	D003	43800.0
E010	Frank	IT	67000.0
E011	Rickard	Hardware	70000.0
E012	Roberto	Hardware	70100.0
E013	Sakshi	IT	67800.0
E014	Eeshan	D001	45600.0
E015	Rajveer	D002	71200.0
E016	Jagdish	D003	46800.0

Another example:

```
package MyJDBCOne;
import java.sql.CallableStatement;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.Scanner;
import oracle.jdbc.OracleTypes;
public class StoredProcedureDemo3 {
    public static void main(String[] args) {
        String dbUrl = "jdbc:oracle:thin:@//localhost:1521/FREE";
        String username = "sys as sysdba";
        String password = "Calcutta1983";
        try {
            Connection con = DriverManager.getConnection(dbUrl, username, password);
            CallableStatement cst = con.prepareCall("call
getallempinfoconditional(?,?)");
            System.out.println("Enter the salary value, to get details of employee
receiving more than this value: ");
            float esal= new java.util.Scanner(System.in).nextFloat();
            cst.setFloat(1, esal);
            cst.registerOutParameter(2, OracleTypes.CURSOR);
            cst.execute();
            ResultSet rs = (ResultSet)cst.getObject(2);
            boolean flag = false;
            System.out.println("Emp_id\tName\t\tdept_id\tbasic_salary");
            System.out.println("-----");
            while (rs.next()) {
                flag = true;
                System.out.println(rs.getString(1) + "\t" +
                    rs.getString(2) + "\t" +
                    rs.getString(3) + "\t\t" +
                    rs.getFloat(4));
            }
        }
    }
}
```

```

        if (!flag) {
            System.out.println("No records available");
        }
    } catch (SQLException e) {
        e.printStackTrace();
    }
}
}

```

Output:

Enter the salary value, to get details of employee receiving more than this value:

60000

Emp_id	Name	dept_id	basic_salary
E002	Naveen Singh	D001	67800.0
E006	Gargi	IT	70000.0
E010	Frank	IT	67000.0
E011	Rickard	Hardware	70000.0
E012	Roberto	Hardware	70100.0
E013	Sakshi	IT	67800.0
E015	Rajveer	D002	71200.0